SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

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QUESTIONS Total Marks: 10

Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	**1**
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	**1**
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	**1**
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	**1**
5	Functional dependency X → Y means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	**1**
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	**1**
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An	BL2 – Understand	**1**

	attribute that depends on a non-prime attribute (d) A foreign key reference		
8	If R(A,B,C) has FD: A→B and B→C, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	**1**
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	**1**
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	**1**

Code of Conduct:

I __K.SANDHYA 192424061_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

MCQ					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts

Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

QUESTION : 1. Which of the following is true about 1NF?

(a) Allows multi-valued attributes
(b) Eliminates partial dependencies
(c) All attributes must be atomic
(d) Eliminates transitive dependencies

Answer: (c) All attributes must be atomic

Reason:

1. 1NF requires every attribute to contain single, indivisible (atomic) values.
2. Multi-valued or repeating groups are not allowed in 1NF.
3. Partial dependencies are removed in 2NF, while transitive dependencies are removed in 3NF.

QUESTION : 2. A relation is in 2NF if it is in 1NF and:

(a) Has no transitive dependencies
(b) Has no partial dependencies on primary key
(c) Every attribute is a prime attribute
(d) Has no multi-valued dependencies

Answer: (b) Has no partial dependencies on primary key

Reason:

1. 2NF must first satisfy all conditions of 1NF.
2. It eliminates partial dependency, where a non-key attribute depends on only part of a composite key.
3. Transitive dependencies are handled by 3NF, not 2NF.

QUESTION : 3. Which normal form deals with multi-valued dependencies?

(a) 2NF
(b) 3NF
(c) BCNF
(d) 4NF

Answer: (d) 4NF

Reason:

1. 4NF specifically deals with multi-valued dependencies.
2. It requires that non-trivial multi-valued dependencies have a suitable decomposition.

3. 4NF helps eliminate redundancy caused by independent multi-valued facts.

QUESTION : 4. BCNF is a stricter version of:

(a) 1NF
(b) 2NF
(c) 3NF
(d) 4NF

Answer: (c) 3NF

Reason:

1. BCNF stands for Boyce-Codd Normal Form.
2. BCNF is stricter than 3NF.
3. Every relation in BCNF satisfies 3NF, but some 3NF relations may not satisfy BCNF.

QUESTION : 5. Functional dependency X → Y means:

(a) Y determines X
(b) X determines Y
(c) X and Y are unrelated
(d) Both determine each other

Answer: (b) X determines Y

Reason:

1. X → Y means the value of X uniquely determines the value of Y.
2. If two tuples have the same X value, they must have the same Y value.
3. X is called the determinant, and Y is the dependent attribute.

QUEATION : 6. Which of the following anomaly is NOT associated with unnormalized relations?

(a) Insertion anomaly
(b) Deletion anomaly
(c) Update anomaly
(d) Retrieval anomaly

Answer: (d) Retrieval anomaly

Reason:

1. Insertion, deletion, and update anomalies are common problems caused by poor database design.
2. Normalization is mainly used to eliminate these three anomalies.
3. Retrieval problems are not normally classified as the standard normalization anomalies.

QUESTION : 7. A superkey is:

(a) A minimal set of attributes that uniquely identifies tuples
(b) Any set of attributes that uniquely identifies tuples
(c) An attribute that depends on a non-prime attribute
(d) A foreign key reference

Answer: (b) Any set of attributes that uniquely identifies tuples

Reason:

1. A superkey is any combination of attributes that uniquely identifies each tuple.
2. It may contain extra attributes that are not necessary.
3. A candidate key is a minimal superkey, so option (a) describes a candidate key.

QUESTION : 8. If R(A,B,C) has FD: A → B and B → C, then C is:

(a) Partially dependent on A
(b) Fully dependent on A
(c) Transitively dependent on A
(d) Independently determined

Answer: (c) Transitively dependent on A

Reason:

1. We have A → B and B → C.
2. Therefore, A determines C indirectly through B.
3. This is called a transitive dependency: A → B → C.

QUESTION : 9. 5NF is also known as:

(a) Domain-Key Normal Form
(b) Boyce-Codd Normal Form
(c) Project-Join Normal Form
(d) Element Normal Form

Answer: (c) Project-Join Normal Form

Reason:

1. 5NF stands for Fifth Normal Form.
2. It is also called Project-Join Normal Form (PJ/NF).
3. It deals with join dependencies and ensures that a relation cannot be further decomposed without losing information.

QUESTION : 10. Lossless-join decomposition ensures:

(a) No data is lost during decomposition
(b) All FDs are preserved
(c) The relation is in BCNF
(d) No null values exist

Answer: (a) No data is lost during decomposition

Reason:

1. A lossless decomposition allows the original relation to be reconstructed by joining the decomposed relations.
2. It ensures that no information is lost during decomposition.
3. Lossless join does not necessarily guarantee dependency preservation or BCNF.